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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

Pharmacy Medicine Stock and Billing Management System

Submitted by

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Semester: **I**

DECLARATION

I hereby declare that the project entitled "**Pharmacy Medicine Stock and Billing Management System "** submitted in partial fulfillment of the requirements for the course **[Java programming]** is a **record of my original work** carried out under the guidance of **[Aravind]**, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

KARTHIGA DEVI .D

(Your Name)

CERTIFICATE

This is to certify that the project entitled **"Pharmacy Medicine Stock and Billing Management System "** is a **bona fide work** carried out by **[KARTHIGA DEVI D] [1BCS CS -B]**, [BSC.CS], **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **[java programming]**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: Coimbatore

Date: 10/04/2026

Submitted for the Viva-Voce Examination held on _____

Guide: **Head of the Department**(Technical Trainer, KGCAS) (Technical Head, KGCAS)

_____ _____

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ABSTRACT

The **Pharmacy Medicine Stock and Billing Management System** is a software application developed to automate the management of medicines, stock control, and billing operations in a pharmacy. In many small and medium pharmacies, stock handling and billing are done manually, which can lead to errors, delays, and difficulty in maintaining records.

The system allows users to select food items from a predefined menu, enter the quantity of each item, and automatically calculate the total bill including applicable taxes such as GST. The application is developed using **Core Java programming concepts**, including classes, objects, methods, conditional statements, loops, and exception handling. The project can be implemented either as a **console-based application or using Java Swing for a graphical user interface**.

The main objective of the system is to improve billing efficiency, reduce human errors, and provide a faster service to customers. The system generates a clear and structured bill that displays the ordered items, quantities, prices, subtotal, tax, and final payable amount.

This project demonstrates the practical implementation of **object-oriented programming concepts in Java** and provides a basic foundation for developing more advanced restaurant management systems in the future.

1. INTRODUCTION

In today's fast-paced world, the healthcare sector plays a crucial role in ensuring the well-being of people. Pharmacies are an essential part of this system, as they are responsible for dispensing medicines and healthcare products to patients. Efficient management of pharmacy operations, especially medicine stock and billing, is necessary to provide quick and accurate service to customers.

Traditionally, many small and medium-scale pharmacies rely on manual methods for maintaining records and generating bills. These methods include handwritten registers, notebooks, or simple calculators. While such systems may work for small volumes, they become inefficient as the number of medicines and customers increases. Manual systems are time-consuming and highly prone to human errors, which can lead to incorrect billing, improper stock tracking, and poor record management.

One of the major challenges in manual pharmacy management is maintaining accurate stock information. Pharmacists must keep track of hundreds of medicines, each with different quantities and expiry dates. Without an automated system, it becomes difficult to monitor stock levels effectively. This may result in overstocking, shortage of essential medicines, or even the sale of expired products, which can pose serious health risks.

Another important issue is the billing process. Manual billing requires the pharmacist to calculate the total cost of medicines using a calculator or by hand. This process is not only slow but also increases the chances of calculation errors. Incorrect billing can lead to financial loss for the pharmacy and dissatisfaction among customers. In busy situations, such delays can reduce the overall efficiency of the pharmacy.

1.1 Objective

The primary objective of this project is to develop a reliable and efficient system for managing pharmacy stock and billing operations.

Detailed Objectives:

- To design and develop a computerized pharmacy management system
- To automate the process of stock management
- To generate accurate and quick billing
- To reduce manual errors in calculations
- To maintain proper records of medicines and sales
- To track expiry dates and avoid selling expired medicines
- To improve customer service by reducing waiting time
- To demonstrate the use of Core Java concepts in real-world applications

1.2 Overview

The Pharmacy Medicine Stock and Billing Management System is a menu-driven application developed using Java. The system provides a simple and user-friendly interface for managing pharmacy operations.

The application allows the user to add new medicines, update stock details, check availability, and generate bills. All operations are performed efficiently using programming logic.

The system stores medicine information and updates stock automatically after each sale. During billing, the system calculates the total price based on quantity and applies tax to generate the final amount.

This system reduces manual effort, ensures accuracy, and improves the overall performance of pharmacy management.

1.3 Features

The system includes the following important features:

- The system allows adding new medicines with details like name, price, quantity, and expiry date.
- It helps in updating and deleting existing medicine records easily.
- The system maintains accurate stock information of all medicines.
- It automatically updates the stock after each sale.
- The system tracks expiry dates of medicines to avoid selling expired products.
- It provides an easy and fast billing system for customers.
- The system calculates the total amount automatically.
- It includes automatic GST/tax calculation in billing.
- The system generates a clear and structured bill (invoice).
- It provides a simple and user-friendly menu-driven interface.
- The system stores data permanently using file handling.
- It retrieves saved data when the program is restarted.
- The system handles errors using exception handling.
- It reduces manual work and improves efficiency.
- The system saves time and minimizes calculation errors.

1. PROGRAM ANALYSIS

Program analysis is an important phase in software development where the problem is studied in detail and the requirements of the system are clearly identified. In the Pharmacy Medicine Stock and Billing Management System, this phase helps in understanding the difficulties faced in manual stock management and billing processes. It focuses on identifying issues such as incorrect billing, poor stock tracking, and lack of expiry monitoring in existing systems. Based on this analysis, the necessary features like medicine management, stock control, and automated billing are defined. This stage ensures that the developed system provides an efficient, accurate, and user-friendly solution for pharmacy operations.

2.1 Problem Definition

In many small pharmacies and medical stores, stock management and billing are often performed manually using registers or simple calculators. This traditional method creates several challenges.

Some common problems include:
• Errors in bill calculation
• Difficulty in maintaining accurate stock records
• Lack of proper tracking of medicine expiry dates
• Time-consuming billing process
• Poor record management of sales and purchases

Manual systems also increase the workload for pharmacists and may lead to serious issues such as selling expired medicines or running out of essential stock. This can affect both business efficiency and customer safety.

Therefore, there is a need for an automated system that can efficiently manage medicine stock, track expiry dates, and generate accurate bills.

The Pharmacy Medicine Stock and Billing Management System is designed to solve these problems by providing a computerized solution that automates stock management and billing processes, ensuring accuracy, efficiency, and better service.

2.2 Project Overview

The Pharmacy Medicine Stock and Billing Management System is a Java-based application developed to simplify and automate the management of medicine stock and billing in pharmacies.

The system provides a menu-driven interface where users can add medicines, view available stock, and generate bills based on customer purchases. The system automatically calculates the total cost of selected medicines and generates an accurate bill.

The system includes the following functionalities:
• Adding and managing medicine details such as name, price, quantity, and expiry date
• Displaying available medicines and stock information
• Selecting medicines and entering quantity for billing
• Automatically calculating the total cost of medicines
• Computing subtotal and tax (GST) amount
• Generating the final bill for the customer

This project is implemented using Core Java programming concepts and can be developed as a console-based application or can be extended into a Java Swing GUI application for better user interaction.

2.3 Scopes and Limitations

Scope

The scope of the Pharmacy Medicine Stock and Billing Management System includes the following:

• Managing medicine details such as name, price, quantity, and expiry date
• Maintaining accurate stock records of all medicines
• Updating stock automatically after each transaction
• Tracking expiry dates to avoid selling expired medicines
• Processing customer purchases efficiently
• Calculating the total bill automatically
• Computing GST/tax and including it in the final bill
• Generating a clear and detailed invoice for customers
• Reducing manual errors in billing and stock management
• Saving and retrieving data using file handling
• Providing a simple and user-friendly interface
• Improving the efficiency and speed of pharmacy operations
• Demonstrating the practical use of Core Java programming concepts

Limitation:

The scope of the Pharmacy Medicine Stock and Billing Management System includes the following:

• Managing medicine details such as name, price, quantity, and expiry date
• Maintaining accurate stock records of all medicines
• Updating stock automatically after each transaction
• Tracking expiry dates to avoid selling expired medicines
• Processing customer purchases efficiently
• Calculating the total bill automatically
• Computing GST/tax and including it in the final bill
• Generating a clear and detailed invoice for customers
• Reducing manual errors in billing and stock management
• Saving and retrieving data using file handling
• Providing a simple and user-friendly interface
• Improving the efficiency and speed of phamacy operation.

1. SYSTEM DESIGN

System design is the stage where the structure and components of the system are planned before the actual implementation. In this phase, the system is divided into different modules, and the workflow of the application is clearly defined. Proper system design helps in organizing the program efficiently and makes development easier and more systematic.

The Pharmacy Medicine Stock and Billing Management System is designed as a menu-driven application where users can add medicines, view stock details, and generate bills automatically based on customer purchases. The system ensures smooth handling of pharmacy operations with accuracy and speed.

The system is divided into several modules to perform specific tasks. Each module is responsible for handling a particular function such as managing medicine details, tracking stock, processing billing, and storing data. This modular design improves the clarity, maintainability, and efficiency of the system.

3.1 Modules

The main modules of the Pharmacy Medicine Stock and Billing Management System are explained below:

1. Medicine Management Module

This module is responsible for managing all medicine-related details such as name, price, quantity, and expiry date. It allows the user to add new medicines and maintain updated records.

Functions of this module include:
• Adding new medicines to the system
• Updating existing medicine details
• Deleting medicines if required
• Storing medicine information

This module ensures that all medicine data is organized and easily accessible.

2. Stock Display Module

This module displays the list of available medicines along with their details such as price, quantity, and expiry date.

Example Display:

1. Paracetamol ₹20 Qty: 50 Exp: 12/2026
2. Amoxicillin ₹50 Qty: 30 Exp: 10/2025
3. Cough Syrup ₹80 Qty: 20 Exp: 08/2025

The stock is displayed so that the user can easily view available medicines before billing.

3. Billing Module

This module allows the user to select medicines and enter the required quantity for generating a bill.

Functions of this module include:
- Selecting medicines from the list
- Entering quantity required
- Checking availability of stock
- Storing selected items for billing

The system allows multiple medicines to be selected before generating the final bill.

4. Bill Calculation Module

This module performs all billing-related calculations automatically.

The system calculates:
- Total cost of each medicine
- Subtotal of all selected medicines
- Applicable tax (GST)
- Final payable amount

The calculation is done using Java programming logic.

Example Calculation:
Item Total = Price × Quantity
GST = 5% of Subtotal
Final Bill = Subtotal + GST

5. Invoice Generation Module

This module generates the final bill for the customer.

The bill includes:
- Medicine name
- Quantity

- Price
- Subtotal
- GST amount
- Final bill amount

Example Bill Output:

------ PHARMACY BILL ------

Item	Qty	Price
Paracetamol	2	40
Cough Syrup	1	80
Subtotal		120
GST (5%)		6
Total		126

Thank You! Visit Again!

6. Exit Module

This module allows the user to exit the application safely after completing all operations.

When the user selects the exit option, the system saves data (if required) and terminates the program.

1. METHODOLOGY

Methodology describes the systematic process followed to develop the *Pharmacy Medicine Stock and Billing Management System*, ensuring efficient and accurate management of pharmacy operations. The development began with problem analysis, where issues in the manual system such as stock mismanagement, billing errors, and difficulty in tracking expiry dates were identified. In the system design phase, the project was structured into modules like medicine management, stock control, billing, and expiry tracking, with simple and user-friendly design planning. During program development, the system was implemented using C language with functions for adding medicines, updating stock, searching records, and generating bills. The system was then tested and validated using various inputs to ensure correctness, reliability, and error-free performance. Finally, the results were documented with program details, sample outputs, and explanations, making the system easy to understand and maintain.

4.1. Problem Analysis

In this phase, the problem is studied and analyzed in detail. The requirements of the system are identified based on the needs of pharmacy operations.

The main problems identified in manual pharmacy management systems are:
- Manual stock maintenance may lead to incorrect quantity records.
- Difficulty in tracking expiry dates of medicines.
- Billing calculations may lead to errors.
- Preparing bills manually consumes more time.
- Difficult to search and update medicine details quickly.
- Lack of proper records for sales and stock transactions.

Based on these problems, the system requirements were defined to develop an automated application that efficiently manages medicine stock, tracks expiry dates, and generates accurate bills, thereby simplifying pharmacy operations.

4.2. System Design

In this stage, the overall structure of the system is designed before coding begins.

The system is divided into modules such as:
- Medicine Management Module
- Stock Management Module
- Billing Module
- Expiry Tracking Module
- Search and Update Module
- Exit Module

The program flow and structure are designed in a simple and user-friendly manner so that the user can easily add, update, search medicines, manage stock, and generate bills efficiently.

4.3. Program Development

After designing the system, the program is implemented using the C programming language.

The following steps were followed during development:
- Creation of functions for different modules
- Implementation of menu-driven logic
- Writing code for medicine stock management
- Developing billing calculation functionality
- Handling user input and validation
- Displaying medicine details and final bill

Core C concepts such as functions, arrays, structures, loops, conditional statements, and file handling were used during development to ensure efficient and accurate system performance.

4.4. Testing and Validation

Testing ensures that the program works correctly and produces the expected results. Different test cases were used to check the functionality of the system.

Example Test Cases:

Test Case	Expected Result
Add new medicine	Medicine details stored correctly
Search Medicine	Correct medicine information displayed
Update stock	Stock updated Accurately
Generate bill	Error handled properly
Enter invalid input	Error handled properly

This phase ensures that the program runs without errors, maintains accurate stock records, and generates correct billing results efficiently.

4.5. Result and Documentation

After successful testing, the results of the system are documented. The system manages medicine stock efficiently and generates the final bill based on the selected medicines and quantities.

The results are demonstrated through program execution and screenshots showing the input and output of the system, including adding medicines, updating stock, and billing details.

Proper documentation is prepared to clearly explain the system design, implementation, modules, and overall working of the application, making it easy to understand and use.

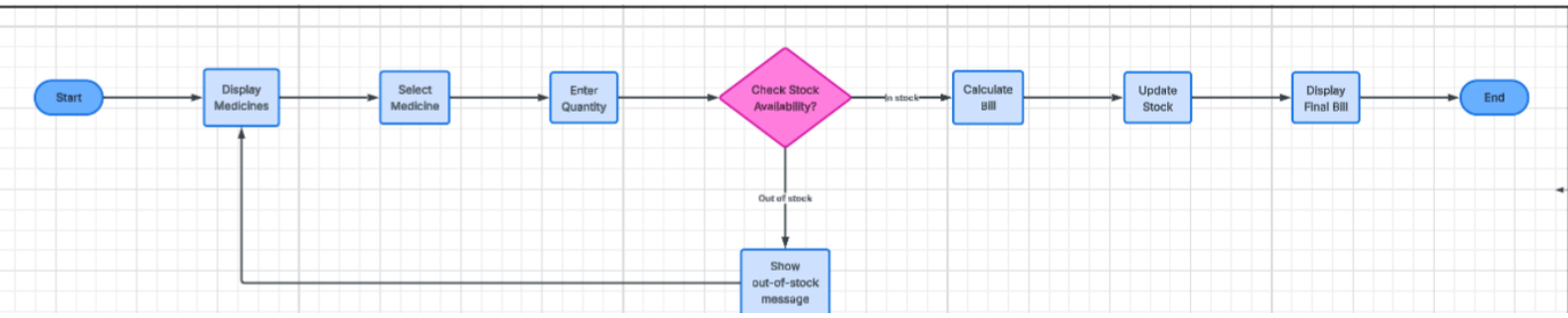
4.6. Algorithm

Algorithm for Pharmacy Medicine Stock and Billing Management System

1. Start the program.

2. Display the list of available medicines with their prices.
3. Ask the user to select a medicine.
4. Enter the required quantity of the selected medicine.
5. Check the availability of stock.
6. Calculate the cost of the selected medicine.
7. Add the cost to the total bill.
8. Update the medicine stock after purchase.
9. Repeat steps 3–8 for additional medicines if required.
10. Calculate the total bill amount.
11. Display the final bill to the user.
12. Stop the program.

4.7. Flowchart



SOURCE CODE

```
import java.util.*;

public class Medicine {

    private String name;

    private double price;

    private int quantity;

    // Constructor

    public Medicine(String name, double price, int quantity) {

        this.name = name;

        this.price = price;
```

```
    this.quantity = quantity;
}

// Getters
public String getName() { return name; }
public double getPrice() { return price; }
public int getQuantity() { return quantity; }

public void setQuantity(int quantity) {
    this.quantity = quantity;
}

public void display() {
    System.out.println(name + " | ₹" + price + " | Qty: " + quantity);
}

//      MAIN METHOD INSIDE MEDICINE (FIXES YOUR ERROR)
public static void main(String[] args) {

    ArrayList<Medicine> list = new ArrayList<>();
    Scanner sc = new Scanner(System.in);

    int choice;

    do {
        System.out.println("\n===== PHARMACY SYSTEM =====");
        System.out.println("1. Add Medicine");
        System.out.println("2. View Stock");
        System.out.println("3. Generate Bill");
        System.out.println("4. Exit");
```

```
System.out.print("Enter choice: ");

choice = sc.nextInt();

switch (choice) {

    case 1:
        sc.nextLine();
        System.out.print("Enter medicine name: ");
        String name = sc.nextLine();

        System.out.print("Enter price: ");
        double price = sc.nextDouble();

        System.out.print("Enter quantity: ");
        int qty = sc.nextInt();

        list.add(new Medicine(name, price, qty));
        System.out.println("     Medicine added!");
        break;

    case 2:
        if (list.isEmpty()) {
            System.out.println("⚠ No medicines available!");
        } else {
            for (int i = 0; i < list.size(); i++) {
                System.out.print((i + 1) + ". ");
                list.get(i).display();
            }
        }
```

```
    break;

case 3:
    if (list.isEmpty()) {
        System.out.println("⚠ No medicines available!");
        break;
    }

    double total = 0;

    for (int i = 0; i < list.size(); i++) {
        System.out.println((i + 1) + ". " + list.get(i).getName());
    }

    System.out.print("Enter number of items: ");
    int n = sc.nextInt();

    for (int i = 0; i < n; i++) {

        System.out.print("Select medicine number: ");
        int index = sc.nextInt() - 1;

        if (index < 0 || index >= list.size()) {
            System.out.println("Invalid!");
            i--;
            continue;
        }

        Medicine m = list.get(index);
```

```
            System.out.print("Enter quantity: ");
            int q = sc.nextInt();

            if (q > m.getQuantity()) {
                System.out.println("Not enough stock!");
                i--;
                continue;
            }

            double cost = q * m.getPrice();
            total += cost;

            m.setQuantity(m.getQuantity() - q);
        }

        double gst = total * 0.05;

        System.out.println("\nTotal: ₹" + total);
        System.out.println("GST: ₹" + gst);
        System.out.println("Final Amount: ₹" + (total + gst));
        break;

    case 4:
        System.out.println("Exiting...");
        break;

    default:
        System.out.println("Invalid choice!");
}
```

```
        } while (choice != 4);
    }
}
```

1. INPUT AND OUTPUT SCREENS

```
===== PHARMACY SYSTEM =====
1. Add Medicine
2. View Stock
3. Generate Bill
4. Exit
Enter choice:
```

input 1:

```
Enter choice: 1
Enter medicine name: paracetamol
Enter price: 40
Enter quantity: 500
? Medicine added!
```

input 2:

```
Enter choice: 2
1. paracetmol | ?40.0 | Qty: 500
```

input 3:

```
Enter choice: 3
1. parectamol
Enter number of items: 1
Select medicine number: 1
Enter quantity: 2

Total: ?80.0
GST: ?4.0
Final Amount: ?84.0
```

input 4:

```
Enter choice: 4
Exiting...
```

1. DISCUSSION & CONCLUSION

7.1 Discussion

The **Pharmacy Medicine Stock and Billing Management System** was developed to automate the process of managing medicine stock and billing in pharmacies. The system allows users to add medicines, update stock details, search for medicines, and generate bills automatically.

During the development process, core C programming concepts such as functions, arrays, structures, loops, and conditional statements were applied. The system efficiently manages medicine records, tracks stock availability, and performs accurate billing operations.

The system was tested with different inputs to ensure that it performs all operations correctly, including stock updates and bill generation. The results show that the system reduces manual effort, minimizes errors, and improves efficiency in pharmacy management.

7.2 Conclusion

The **Pharmacy Medicine Stock and Billing Management System** was developed to automate the process of managing medicine stock and billing in pharmacies. The system allows users to add medicines, update stock, search for medicines, and generate bills automatically.

During the development process, core C programming concepts such as functions, arrays, structures, loops, and conditional statements were applied. The system successfully manages stock details, tracks medicine availability, and performs accurate billing operations.

The system was tested with different inputs to ensure that it works correctly and produces accurate results. The results show that the system reduces manual effort, minimizes errors in stock management and billing, and improves overall efficiency in pharmacy operations.

1. REFERENCES

The following resources were used during the development of this project:

Books

• Herbert Schildt – C: The Complete Reference
• E. Balagurusamy – Programming in ANSI C

Websites

• GeeksforGeeks
• W3Schools
• TutorialsPoint
• Javatpoint